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Barium peroxide

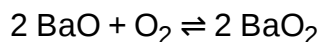
Barium peroxide is an inorganic compound with the formula BaO2. This white solid (gray when impure) is one of the most common inorganic peroxides, and it was the first peroxide compound discovered. Being an oxidizer and giving a vivid green colour upon ignition (as do all barium compounds), it finds some use in fireworks; historically, it was also used as a precursor for hydrogen peroxide.^[3]

Structure

Barium peroxide is a peroxide, containing O_2^{2-} subunits. The solid is isomorphous to calcium carbide, CaC2.

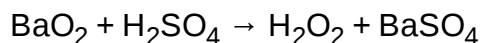
Preparation and use

Barium peroxide arises by the reversible reaction of O_2 with barium oxide. The peroxide forms around 500 °C and oxygen is released above 820 °C.^[1]



This reaction is the basis for the now-obsolete Brin process for separating oxygen from the atmosphere. Other oxides, e.g. Na2O and SrO, behave similarly.^[4]

In another obsolete application, barium peroxide was once used to produce hydrogen peroxide via its reaction with sulfuric acid:^[3]

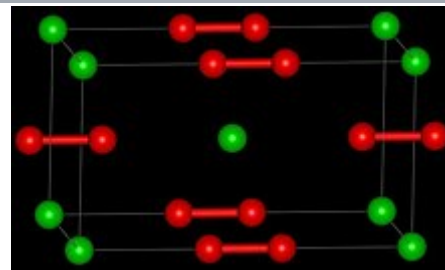


The insoluble barium sulfate is filtered from the mixture.

Footnotes

- Accommodation of Excess Oxygen in Group II Monoxides - S.C. Middleburgh, R.W. Grimes and K.P.D. Lagerlof Journal of the American Ceramic Society 2013, Volume 96, pages 308–311. doi:[10.1111/j.1551-2916.2012.05452.x](https://doi.org/10.1111/j.1551-2916.2012.05452.x) (<https://doi.org/10.1111%2Fj.1551-2916.2012.05452.x>)
- Massalimov, I. A.; Kireeva, M. S.; Sangalov, Yu. A. (2002). "Structure and Properties of Mechanically

Barium peroxide



Names

IUPAC name

barium peroxide

Other names

Barium binoxide,
Barium dioxide

Identifiers

CAS Number

1304-29-6 (https://commonchemistry.cas.org/detail?cas_rn=1304-29-6) ✓

3D model (JSmol)

Interactive image (<https://chemapps.stolaf.edu/jmol/jmol.php?model=%5BBa%2B2%5D.%5BO-%5D>)

ChemSpider

14090 (<https://www.chemspider.com/Chemical-Structure.14090.html>) ✓

Activated Barium Peroxide". *Inorganic Materials*. **38** (4): 363–366. doi:10.1023/A:1015105922260 (<https://doi.org/10.1023%2FA%3A1015105922260>).

- Harald Jakob; Stefan Leininger; Thomas Lehmann; Sylvia Jacobi; Sven Gutewort. "Peroxo Compounds, Inorganic". *Ullmann's Encyclopedia of Industrial Chemistry*. Weinheim: Wiley-VCH. doi:10.1002/14356007.a19_177.pub2 (https://doi.org/10.1002%2F14356007.a19_177.pub2).
- Holleman, A. F.; Wiberg, E. "Inorganic Chemistry" Academic Press: San Diego, 2001. ISBN 0-12-352651-5.

See also

- Barium oxide

External links

- International Chemical Safety Card 0381 (https://www.ilo.org/dyn/icsc/showcard.display?p_lang=en&p_card_id=0381&p_version=2)
- MSDS at sigma aldrich (<http://www.sigmaaldrich.com/MSDS/MSDS/DisplayMSDSPage.do?country=US&language=en&productNumber=11428&brand=SI&PageToGoToURL=http%3A%2F%2Fwww.sigmaaldrich.com%2Fcatalog%2Fsearch%3Finterface%3DAll%26term%3Dbarium%2520peroxide%26lang%3Den%26region%3DUS%26focus%3Dproduct%26N%3D0%2B220003048%2B219853269%2B219853286>)

<u>ECHA InfoCard</u>	100.013.754 (https://echa.europa.eu/substance-information/-/substanceinfo/100.013.754)
<u>EC Number</u>	215-128-4
<u>PubChem CID</u>	14773 (https://pubchem.ncbi.nlm.nih.gov/compound/14773)
<u>RTECS number</u>	CR0175000
<u>UNII</u>	T892KY013Y (https://precision.fda.gov/uniisearch/srs/unii/T892KY013Y) ✓
<u>UN number</u>	1449
<u>CompTox Dashboard (EPA)</u>	DTXSID9051294 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID9051294)
<u>InChI</u>	InChI=1S/Ba.O2/c;1-2/q+2;-2 ✓ Key: ZJRXSAYFZMGQFP-UHFFFAOYSA-N ✓ InChI=1/Ba.O2/c;1-2/q+2;-2 Key: ZJRXSAYFZMGQFP-UHFFFAOYAZ
<u>SMILES</u>	[Ba+2].[O-][O-]
Properties	
<u>Chemical formula</u>	BaO ₂
<u>Molar mass</u>	169.33 g/mol (anhydrous) 313.45 (octahydrate)
<u>Appearance</u>	Grey-white crystalline (anhydrous)

	colorless solid (octahydrate)
<u>Odor</u>	odorless
<u>Density</u>	5.68 g/cm ³ (anhydrous) 2.292 g/cm ³ (octahydrate)
<u>Melting point</u>	450 °C (842 °F; 723 K)
<u>Boiling point</u>	800 °C (1,470 °F; 1,070 K) (decomposes to BaO & O ₂ . ^[1])
<u>Solubility in water</u>	<i>anhydrous</i> 0.091 g/100 mL (20 °C) <i>octahydrate</i> 0.168 g/cm ³
<u>Solubility</u>	dissolves with decomposition in acid
<u>Magnetic susceptibility (χ)</u>	-40.6·10 ⁻⁶ cm ³ /mol
Structure	
<u>Crystal structure</u>	Tetragonal ^[2]
<u>Space group</u>	D ¹⁷ _{4h} , I4/mmm, <u>tl6</u>
<u>Coordination geometry</u>	6
Hazards	
GHS labelling:	
<u>Pictograms</u>	
<u>Signal word</u>	Warning
<u>Hazard statements</u>	<u>H272</u> , <u>H302</u> , <u>H332</u>
<u>Precautionary statements</u>	<u>P210</u> , <u>P220</u> , <u>P221</u> , <u>P261</u> , <u>P264</u> , <u>P270</u> , <u>P271</u> , <u>P280</u> , <u>P301+P312</u> , <u>P304+P312</u> ,

	P304+P340, P312, P330, P370+P378, P501
NFPA 704 (fire diamond)	
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa). ✓ <u>verify</u> (what is ✗ ?) <u>Infobox references</u>	

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